NILPOTENCE TRANSFER BETWEEN P AND THE DIVIDED POWER IDEAL

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Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- n!-identity for divided powers. lemma-silly part (1) (same chapter, named). Supplies x^{pN} = (pN)! gamma_{pN}(x) used in the forward direction.

Lemma 0.1. *Let p be a prime number. Let A be a ring, let $I \subset A$ be an ideal, and let γ be a divided power structure on I. Assume p is nilpotent in A/I. Then I is locally nilpotent if and only if p is nilpotent in A.*

Proof. If $p^N = 0$ in A, then for $x \in I$ we have $x^{pN} = (pN)!\gamma_{pN}(x) = 0$ because $(pN)!$ is divisible by p^N. Conversely, assume I is locally nilpotent. We've also assumed that p is nilpotent in A/I, hence $p^r \in I$ for some r, hence p^r nilpotent, hence p nilpotent. □

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.